

## Quadratic functions - Set 1

TMUA practice questions by topic. Specification MM1.3, M4.5, M4.11-M4.14. 12 questions.

Each question has one correct answer. No calculator. Answers and worked solutions are in the separate solutions PDF.

- 1 The equation  $x^2 + (k - 2)x + (k + 1) = 0$  has two distinct real roots. Find the complete set of possible values of  $k$ .
- A**  $0 < k < 8$   
**B**  $-2 < k < 4$   
**C**  $k < -2$  or  $k > 4$   
**D**  $k < 0$  or  $k > 4$   
**E**  $k < 0$  or  $k > 8$   
**F** all real  $k$
- 2 By completing the square, find the minimum value of  $2x^2 - 12x + 23$ .
- A**  $-13$   
**B**  $5$   
**C**  $23$   
**D**  $7$   
**E**  $-5$   
**F**  $3$
- 3 The quadratic  $y = ax^2 + bx + c$  has its vertex at  $(2, -3)$  and passes through  $(0, 5)$ . Find  $a + b + c$ .
- A**  $-1$   
**B**  $7$   
**C**  $1$   
**D**  $5$   
**E**  $-3$   
**F**  $3$

- 4 For what value of  $c$  does the line  $y = 2x + c$  touch the parabola  $y = x^2 - 4x + 7$ ?
- A** 7  
**B** 2  
**C** -2  
**D** -1  
**E** 0  
**F** 5
- 5 How many real solutions does the equation  $x^4 - 5x^2 + 4 = 0$  have?
- A** 2  
**B** 4  
**C** 5  
**D** 0  
**E** 1  
**F** 3
- 6 The equation  $kx^2 + 4x + k = 0$  has no real roots. Find the complete set of values of  $k$ .
- A**  $0 < k < 2$   
**B**  $k > 2$   
**C**  $k < -2$  or  $k > 2$   
**D**  $-2 < k < 2$   
**E**  $k < -2$   
**F**  $-2 < k < 0$  or  $0 < k < 2$
- 7 The quadratic  $x^2 + px + q$  has minimum value  $-4$  and one of its roots is 0. Which of the following could be the quadratic?
- A**  $x^2 - 6x + 8$   
**B**  $x^2 - 2x - 3$   
**C**  $x^2 + 2x - 8$   
**D**  $x^2 - 4x$   
**E**  $x^2 + 6x + 5$

- 8 The curve  $y = x^2 - 2ax + a^2 + a - 6$  has its vertex on the  $x$ -axis. Find the value of  $a$ .
- A 0  
B 6  
C -6  
D 3  
E 2  
F -2
- 9 The roots of  $2x^2 + bx + 18 = 0$  are equal. Find the sum of the possible values of  $b$ .
- A 36  
B 6  
C 0  
D 12  
E 24  
F -12
- 10 Which of the following quadratics has its vertex in the second quadrant (where  $x < 0$  and  $y > 0$ )?
- A  $y = -x^2 - 2x - 3$   
B  $y = -x^2 + 2x + 1$   
C  $y = x^2 - 4x + 5$   
D  $y = x^2 + 6x + 10$   
E  $y = x^2 + 4x + 1$
- 11 The graph of  $y = x^2 + bx + c$  crosses the  $x$ -axis at  $(1, 0)$  and  $(5, 0)$ . What is the  $y$ -coordinate of the vertex?
- A -9  
B 5  
C -4  
D -3  
E 4  
F -5

**12** Given that  $x^2 + y^2 = 10$  and  $x + y = 4$ , find the value of  $xy$ .

**A** 4

**B** 2

**C** 3

**D** 6

**E**  $-3$

**F** 5